INFORMATIKA UMN

UMN
UNIVERSITAS MULTIMEDIA NUSANTARA

IF332
FORMAL LANGUAGES & AUTOMATA THEORY

14
PUSHDOWN AUTOMATA (PDA)

DENNIS GUNAWAN

REVIEW

Greibach Normal Form (GNF):

Greibach Normal Form (GNF)

Conversion of CNF to GNF

OUTLINE

Pushdown Automata (PDA)

Graphical Notation for PDA

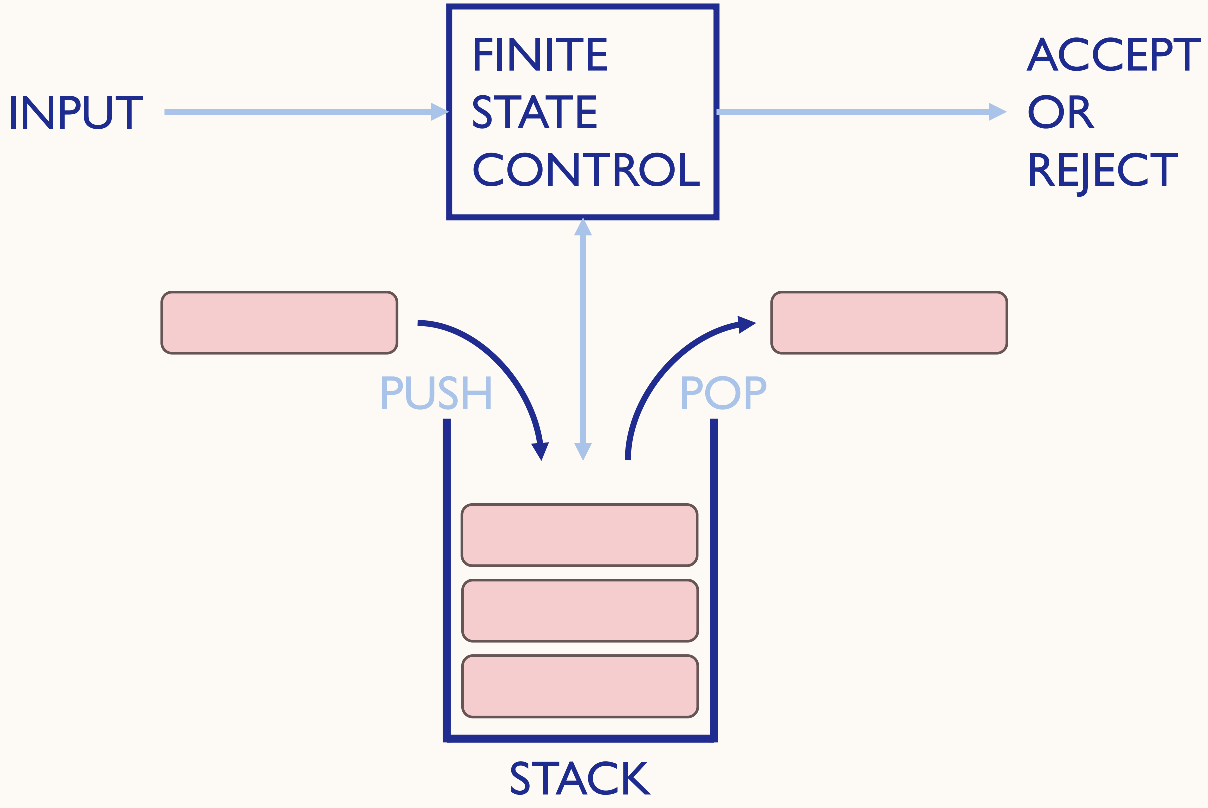
Conversion of CFG to PDA

Instantaneous Descriptions of PDA

PUSHDOWN AUTOMATA (PDA)

- An ϵ-NFA with the addition of a stack
  - The presence of a stack means that the PDA can “remember” an infinite amount of information
  - PDA can only access the information on its stack in a last-in-first-out way
- PDA recognize all and only the context-free languages

PUSHDOWN AUTOMATA (PDA)



PUSHDOWN AUTOMATA (PDA)

- Seven-tuple notation: $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$
  - Q : a finite set of states
  - Σ : a finite set of input symbols
  - Γ : a finite stack alphabet
  - δ : the transition function
  - q_0 : the start state
  - Z_0 : the start stack symbol
  - F : the set of accepting states or final states

PUSHDOWN AUTOMATA (PDA)

- δ takes as argument a triple $\delta(q, a, X)$
  - q : a state in Q
  - a : either an input symbol in Σ or $a = \varepsilon$
  - X : a stack symbol, that is, a member of Γ

- The output of δ is a finite set of pairs (p, γ)
  - p : a new state
  - γ : a string of stack symbols that replaces X at the top of the stack
    - If $\gamma = \varepsilon$ then the stack is popped
    - If $\gamma = X$ then the stack is unchanged
    - If $\gamma = YZ$ then X is replaced by Z and Y is pushed onto the stack

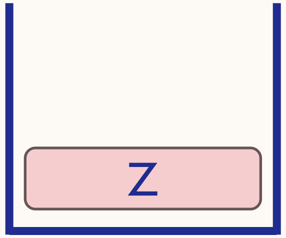
ACCEPTANCE BY FINAL STATE

$P = (\{q_1, q_2\}, \{a, b\}, \{A, B, Z\}, \delta, q_1, Z, \{q_2\})$

<u>Transition function δ</u>:

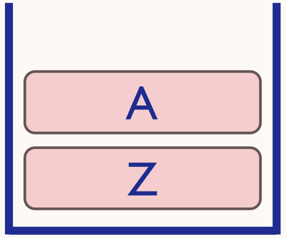
$\delta(q_1, \varepsilon, Z) = (q_2, Z)$
$\delta(q_1, a, Z) = (q_1, AZ)$
$\delta(q_1, b, Z) = (q_1, BZ)$
$\delta(q_1, a, A) = (q_1, AA)$
$\delta(q_1, b, A) = (q_1, \varepsilon)$
$\delta(q_1, a, B) = (q_1, \varepsilon)$
$\delta(q_1, b, B) = (q_1, BB)$

<u>Stack</u>:

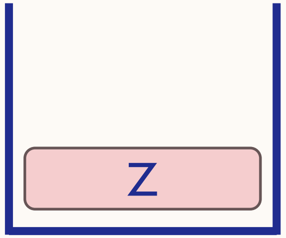


<u>String *abba*</u>:

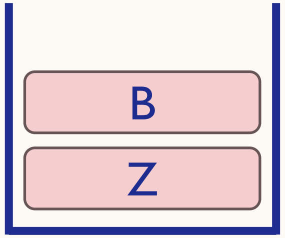
1. $\delta(q_1, a, Z) = (q_1, AZ)$



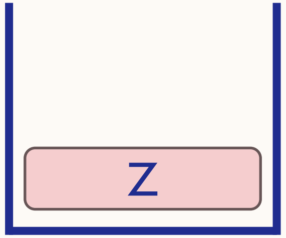
2. $\delta(q_1, b, A) = (q_1, \varepsilon)$



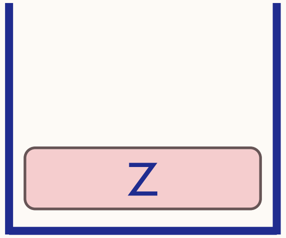
3. $\delta(q_1, b, Z) = (q_1, BZ)$



4. $\delta(q_1, a, B) = (q_1, \varepsilon)$



5. $\delta(q_1, \varepsilon, Z) = (q_2, Z)$



ACCEPTANCE BY EMPTY STACK

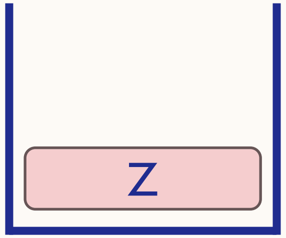
$P = (\{q_1, q_2\}, \{0,1,2\}, \{Z, B, G\}, \delta, q_1, Z, \emptyset)$

Transition function δ:

$\delta(q_1, 0, Z) = (q_1, BZ)$
$\delta(q_1, 0, B) = (q_1, BB)$
$\delta(q_1, 0, G) = (q_1, BG)$
$\delta(q_1, 2, Z) = (q_2, Z)$
$\delta(q_1, 2, B) = (q_2, B)$
$\delta(q_1, 2, G) = (q_2, G)$

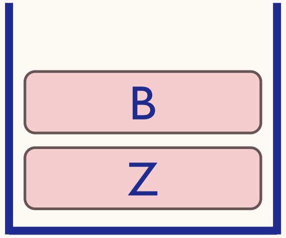
$\delta(q_2, 0, B) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_2, \varepsilon)$
$\delta(q_1, 1, Z) = (q_1, GZ)$
$\delta(q_1, 1, B) = (q_1, GB)$
$\delta(q_1, 1, G) = (q_1, GG)$
$\delta(q_2, 1, G) = (q_2, \varepsilon)$

Stack:

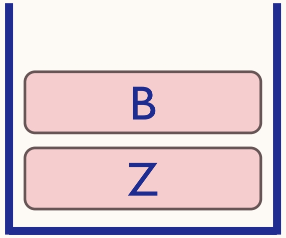


String 020:

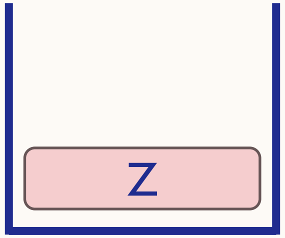
1. $\delta(q_1, 0, Z) = (q_1, BZ)$



2. $\delta(q_1, 2, B) = (q_2, B)$



3. $\delta(q_2, 0, B) = (q_2, \varepsilon)$



4. $\delta(q_2, \varepsilon, Z) = (q_2, \varepsilon)$

GRAPHICAL NOTATION FOR PDA

$L_{wwr} = \{ww^R \mid w \text{ is in } (0+1)^*\}$

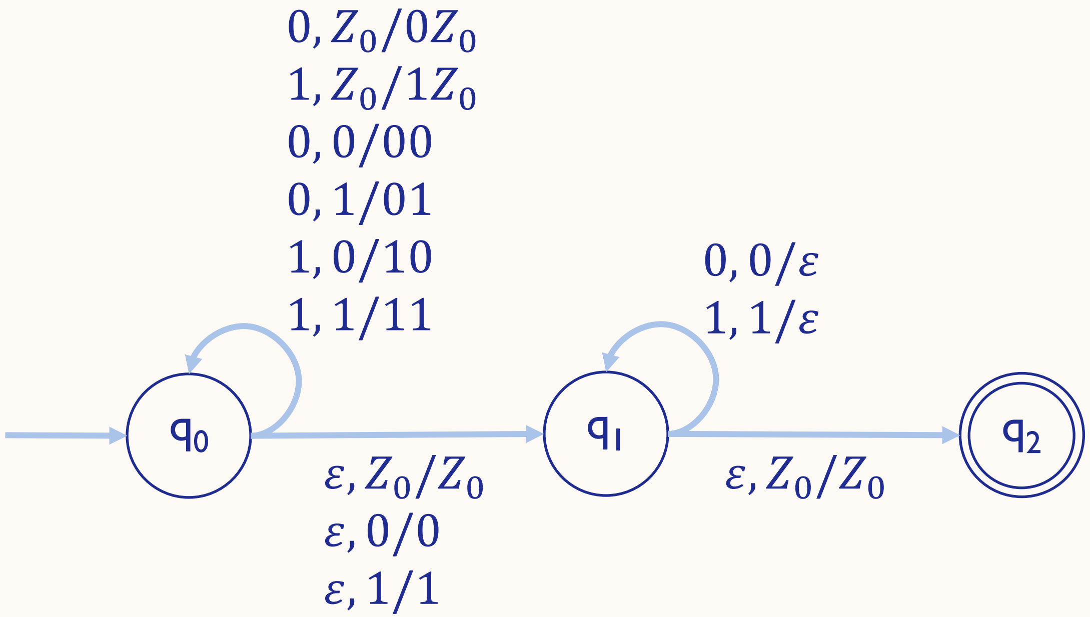
This language, often referred to as "w-w-reversed", is the even-length palindromes over alphabet $\{0,1\}$

$P = (\{q_0, q_1, q_2\}, \{0,1\}, \{0,1,Z_0\}, \delta, q_0, Z_0, \{q_2\})$

Transition function δ:

$\delta(q_0, 0, Z_0) = (q_0, 0Z_0)$
$\delta(q_0, 1, Z_0) = (q_0, 1Z_0)$
$\delta(q_0, 0, 0) = (q_0, 00)$
$\delta(q_0, 0, 1) = (q_0, 01)$
$\delta(q_0, 1, 0) = (q_0, 10)$
$\delta(q_0, 1, 1) = (q_0, 11)$

$\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$
$\delta(q_0, \varepsilon, 0) = (q_1, 0)$
$\delta(q_0, \varepsilon, 1) = (q_1, 1)$
$\delta(q_1, 0, 0) = (q_1, \varepsilon)$
$\delta(q_1, 1, 1) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$



CONVERSION OF CFG TO PDA

- $P = (\{q_1, q_2, q_3\}, \Sigma, \Gamma, \delta, q_1, Z, \{q_3\})$
  - $\Sigma \quad = terminal\ symbols$
  - $\Gamma \quad = variable\ symbols \cup terminal\ symbols \cup Z$

- Transition function δ
  - $\delta(q_1, \varepsilon, Z) = (q_2, SZ)$
  - For each variable $A, \delta(q_2, \varepsilon, A) = \{(q_2, \beta) \mid A \rightarrow \beta\ is\ a\ production\ of\ CFG\}$
  - For each terminal $a, \delta(q_2, a, a) = (q_2, \varepsilon)$
  - $\delta(q_2, \varepsilon, Z) = (q_3, Z)$

CONVERSION OF CFG TO PDA

Convert the CFG below to a PDA that accepts the same language.

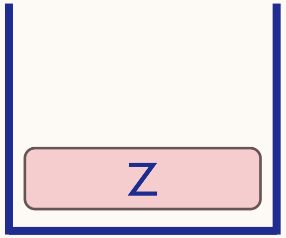
$D \rightarrow aDa \mid bDb \mid c$

$P = (\{q_1, q_2, q_3\}, \{a, b, c\}, \{D, a, b, c, Z\}, \delta, q_1, Z, \{q_3\})$

<u>Transition function δ</u>:

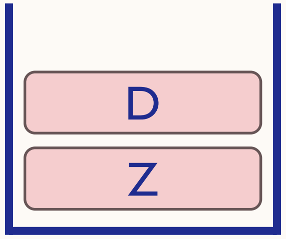
$\delta(q_1, \varepsilon, Z) = (q_2, DZ)$
$\delta(q_2, \varepsilon, D) = \{(q_2, aDa), (q_2, bDb), (q_2, c)\}$
$\delta(q_2, a, a) = \delta(q_2, b, b) = \delta(q_2, c, c) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_3, Z)$

<u>Stack</u>:

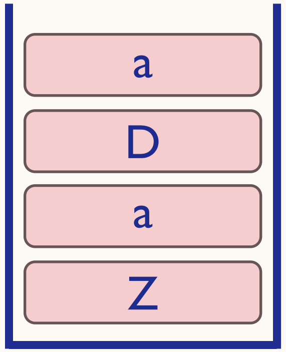


<u>$D \Rightarrow aDa \Rightarrow aca$</u>:

1. $\delta(q_1, \varepsilon, Z) = (q_2, DZ)$



2. $\delta(q_2, \varepsilon, D) = (q_2, aDa)$



CONVERSION OF CFG TO PDA

Convert the CFG below to a PDA that accepts the same language.

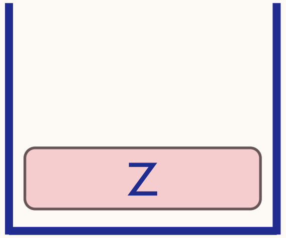
$D \to aDa \mid bDb \mid c$

$P = (\{q_1, q_2, q_3\}, \{a, b, c\}, \{D, a, b, c, Z\}, \delta, q_1, Z, \{q_3\})$

<u>Transition function δ</u>:

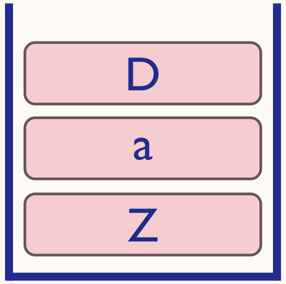
$\delta(q_1, \varepsilon, Z) = (q_2, DZ)$
$\delta(q_2, \varepsilon, D) = \{(q_2, aDa), (q_2, bDb), (q_2, c)\}$
$\delta(q_2, a, a) = \delta(q_2, b, b) = \delta(q_2, c, c) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_3, Z)$

<u>Stack</u>:

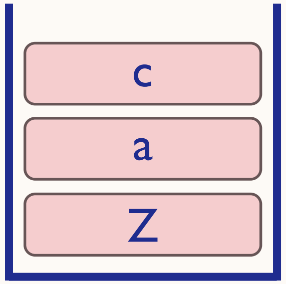


<u>$D \Rightarrow aDa \Rightarrow aca$</u>:

3. $\delta(q_2, a, a) = (q_2, \varepsilon)$



4. $\delta(q_2, \varepsilon, D) = (q_2, c)$



CONVERSION OF CFG TO PDA

Convert the CFG below to a PDA that accepts the same language.

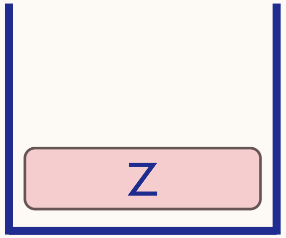
$D \rightarrow aDa \mid bDb \mid c$

$P = (\{q_1, q_2, q_3\}, \{a, b, c\}, \{D, a, b, c, Z\}, \delta, q_1, Z, \{q_3\})$

Transition function δ:

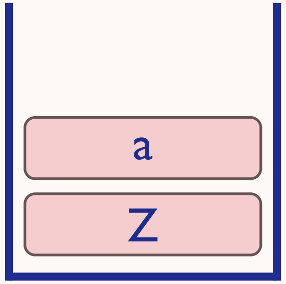
$\delta(q_1, \varepsilon, Z) = (q_2, DZ)$
$\delta(q_2, \varepsilon, D) = \{(q_2, aDa), (q_2, bDb), (q_2, c)\}$
$\delta(q_2, a, a) = \delta(q_2, b, b) = \delta(q_2, c, c) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_3, Z)$

Stack:

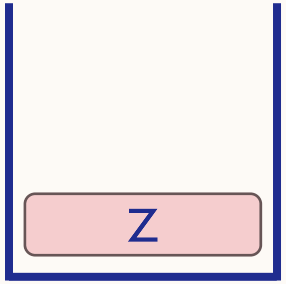


$D \Rightarrow aDa \Rightarrow aca$:

5. $\delta(q_2, c, c) = (q_2, \varepsilon)$



6. $\delta(q_2, a, a) = (q_2, \varepsilon)$



CONVERSION OF CFG TO PDA

Convert the CFG below to a PDA that accepts the same language.

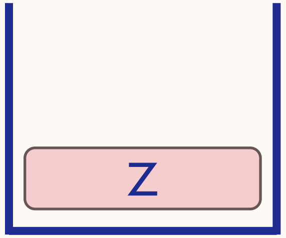
$D \rightarrow aDa \mid bDb \mid c$

$P = (\{q_1, q_2, q_3\}, \{a, b, c\}, \{D, a, b, c, Z\}, \delta, q_1, Z, \{q_3\})$

Transition function δ:

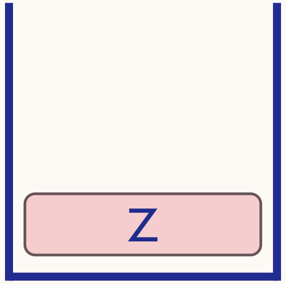
$\delta(q_1, \varepsilon, Z) = (q_2, DZ)$
$\delta(q_2, \varepsilon, D) = \{(q_2, aDa), (q_2, bDb), (q_2, c)\}$
$\delta(q_2, a, a) = \delta(q_2, b, b) = \delta(q_2, c, c) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_3, Z)$

Stack:



$D \Rightarrow aDa \Rightarrow aca$:

7. $\delta(q_2, \varepsilon, Z) = (q_3, Z)$



INSTANTANEOUS DESCRIPTIONS (ID) OF PDA

- A triple (q, w, γ)
  - q : the state
  - w : the remaining input
  - γ : the stack contents

- Suppose $\delta(q, a, X)$ contains (p, α)
  - Then for all strings w in Σ^* and β in Γ^*: $(q, aw, X\beta) \vdash (p, w, \alpha\beta)$

INSTANTANEOUS DESCRIPTIONS (ID) OF PDA

$P = (\{q_1, q_2, q_3\}, \{a, b, c\}, \{D, a, b, c, Z\}, \delta, q_1, Z, \{q_3\})$

Transition function δ: $\delta(q_1, \varepsilon, Z) = (q_2, DZ)$
$\delta(q_2, \varepsilon, D) = \{(q_2, aDa), (q_2, bDb), (q_2, c)\}$
$\delta(q_2, a, a) = \delta(q_2, b, b) = \delta(q_2, c, c) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_3, Z)$

String aca: $(q_1, aca, Z) \vdash (q_2, aca, DZ) \vdash (q_2, aca, aDaZ) \vdash (q_2, ca, DaZ)$
$\vdash (q_2, ca, caZ) \vdash (q_2, a, aZ) \vdash (q_2, \varepsilon, Z) \vdash (q_3, \varepsilon, Z)$

INSTANTANEOUS DESCRIPTIONS (ID) OF PDA

$P = (\{q_0, q_1, q_2\}, \{0,1\}, \{0,1,Z_0\}, \delta, q_0, Z_0, \{q_2\})$

Transition function δ:

$\delta(q_0, 0, Z_0) = (q_0, 0Z_0)$
$\delta(q_0, 1, Z_0) = (q_0, 1Z_0)$
$\delta(q_0, 0, 0) = (q_0, 00)$
$\delta(q_0, 0, 1) = (q_0, 01)$
$\delta(q_0, 1, 0) = (q_0, 10)$
$\delta(q_0, 1, 1) = (q_0, 11)$
$\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$
$\delta(q_0, \varepsilon, 0) = (q_1, 0)$
$\delta(q_0, \varepsilon, 1) = (q_1, 1)$
$\delta(q_1, 0, 0) = (q_1, \varepsilon)$
$\delta(q_1, 1, 1) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$

String 1111: $(q_0, 1111, Z_0) \vdash (q_0, 111, 1Z_0) \vdash (q_0, 11, 11Z_0)$
$\vdash (q_1, 11, 11Z_0) \vdash (q_1, 1, 1Z_0) \vdash (q_1, \varepsilon, Z_0) \vdash (q_2, \varepsilon, Z_0)$

PRACTICE

EXERCISES

1. Suppose the PDA $P = (\{q,p\},\{0,1\},\{Z_0,X\},\delta,q,Z_0,\{p\})$ has the following transition function:

   1. $\delta(q,0,Z_0) = (q,XZ_0)$
   2. $\delta(q,0,X) = (q,XX)$
   3. $\delta(q,1,X) = (q,X)$
   4. $\delta(q,\varepsilon,X) = (p,\varepsilon)$
   5. $\delta(p,\varepsilon,X) = (p,\varepsilon)$
   6. $\delta(p,1,X) = (p,XX)$
   7. $\delta(p,1,Z_0) = (p,\varepsilon)$

   a. Give the transition diagram for the PDA above.

   b. Starting from the initial ID (q,w,Z_0), show all the reachable ID's when the input w is:
   (i) 01 (ii) 0011 (iii) 010

EXERCISES

2. Suppose the PDA $P = (\{q_0, q_1\}, \{0,1\}, \{X, Z\}, \delta, q_0, Z, \emptyset)$ has the following transition function:

   1. $\delta(q_0, 1, Z) = (q_0, XZ)$
   2. $\delta(q_0, 1, X) = (q_0, XX)$
   3. $\delta(q_0, 0, X) = (q_1, X)$
   4. $\delta(q_0, \varepsilon, Z) = (q_0, \varepsilon)$
   5. $\delta(q_1, 1, X) = (q_1, \varepsilon)$
   6. $\delta(q_1, 0, Z) = (q_0, Z)$

   a. Give the transition diagram for the PDA above.

   b. Using sequence of ID's, show whether each of the strings below is accepted or rejected.
   (i) 111011 (ii) 110110 (iii) 001100

EXERCISES

3. PDA $P = (\{q_0, q_1, q_2, q_3, f\}, \{a, b\}, \{Z_0, A, B\}, \delta, q_0, Z_0, \{f\})$ has the following rules defining δ:

1. $\delta(q_0, a, Z_0) = (q_1, AAZ_0)$
2. $\delta(q_0, b, Z_0) = (q_2, BZ_0)$
3. $\delta(q_0, \varepsilon, Z_0) = (f, \varepsilon)$
4. $\delta(q_1, a, A) = (q_1, AAA)$
5. $\delta(q_1, b, A) = (q_1, \varepsilon)$
6. $\delta(q_1, \varepsilon, Z_0) = (q_0, Z_0)$
7. $\delta(q_2, a, B) = (q_3, \varepsilon)$
8. $\delta(q_2, b, B) = (q_2, BB)$
9. $\delta(q_2, \varepsilon, Z_0) = (q_0, Z_0)$
10. $\delta(q_3, \varepsilon, B) = (q_2, \varepsilon)$
11. $\delta(q_3, \varepsilon, Z_0) = (q_1, AZ_0)$

a. Give an execution trace (sequence of ID's) showing that string bab is in $L(P)$.

b. Give an execution trace showing that abb is in $L(P)$.

EXERCISES

3. PDA $P = (\{q_0, q_1, q_2, q_3, f\}, \{a, b\}, \{Z_0, A, B\}, \delta, q_0, Z_0, \{f\})$ has the following rules defining δ:

1. $\delta(q_0, a, Z_0) = (q_1, AAZ_0)$
2. $\delta(q_0, b, Z_0) = (q_2, BZ_0)$
3. $\delta(q_0, \varepsilon, Z_0) = (f, \varepsilon)$
4. $\delta(q_1, a, A) = (q_1, AAA)$
5. $\delta(q_1, b, A) = (q_1, \varepsilon)$
6. $\delta(q_1, \varepsilon, Z_0) = (q_0, Z_0)$
7. $\delta(q_2, a, B) = (q_3, \varepsilon)$
8. $\delta(q_2, b, B) = (q_2, BB)$
9. $\delta(q_2, \varepsilon, Z_0) = (q_0, Z_0)$
10. $\delta(q_3, \varepsilon, B) = (q_2, \varepsilon)$
11. $\delta(q_3, \varepsilon, Z_0) = (q_1, AZ_0)$

c. Give the contents of the stack after P has read b^7a^4 from its input.

EXERCISES

4. Convert the grammar below to a PDA that accepts the same language.

   $S \rightarrow 0S1 \mid A$
   $A \rightarrow 1A0 \mid S \mid \varepsilon$

5. Convert the grammar below to a PDA that accepts the same language.

   $S \rightarrow aAA$
   $A \rightarrow aS \mid bS \mid a$

EXERCISES

6. Design a PDA to accept the following language.

   $\{0^n 1^n \mid n \geq 1\}$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

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To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology

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